

Department of Mathematics, IIT Madras
MA2031-Linear Algebra for Engineers
Assignment-II

- ✓ 1. In each of the following, a vector space V and a subset S of V are given. Determine whether S is linearly dependent; and if it is, express one of the vectors in S as a linear combination of the remaining vectors.
 - ✓ (a) $V = \mathbb{R}^3$, $S = \{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$.
 - ✓ (b) $V = \mathbb{R}^3$, $S = \{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$.
 - ✓ (c) $V = \mathbb{F}_3[t]$, $S = \{t^2 - 3t + 5, t^3 + 2t^2 - t + 1, t^3 + 3t^2 - 1\}$.
 - ✓ (d) $V = \mathbb{F}^{2 \times 2}$, $S = \left\{ \begin{bmatrix} 2 & 0 \\ 2 & 3 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \right\}$.
 - ✓ (e) V = the set of all functions from $\mathbb{R}$ to $\mathbb{R}$, $S = \{2, \sin^2 t, \cos^2 t\}$.
 - ✓ (f) V = the set of all functions from $\mathbb{R}$ to $\mathbb{R}$, $S = \{1, \sin t, \sin 2t\}$.
- ✓ 2. Show that in $\mathbb{R}^2$, $\{(a, b), (c, d)\}$ is linearly independent iff $ad - bc \neq 0$.
- ✓ 3. Show that in $\mathbb{R}^2$, any three vectors are linearly dependent.
- ✓ 4. Give three linearly dependent vectors in $\mathbb{R}^2$ such that none is a scalar multiple of another.
- ✓ 5. Give four vectors in $\mathbb{R}^3$ which are linearly dependent so that any three of them are linearly independent.
- ✓ 6. Construct examples to show that the following statements are false:
 - ✓ (a) Each subset of a linearly dependent set is linearly dependent.
 - ✓ (b) Each superset of a linearly independent set is linearly independent.
- ✓ 7. Show that in the vector space of all functions from $\mathbb{R}$ to $\mathbb{R}$, the set of functions $\{e^t, te^t, t^3 e^t\}$ is linearly independent.
- ✓ 8. Is $\{\sin t, \sin 2t, \sin 3t, \dots, \sin nt\}$ linearly independent in $C[-\pi, \pi]$?
- ✓ 9. Let $p_1(t), \dots, p_k(t)$ be polynomials with coefficients from $\mathbb{F}$ with distinct degrees. Show that $\{p_1(t), \dots, p_k(t)\}$ is linearly independent in $\mathbb{F}[t]$.
- ✓ 10. Determine which of the following sets are bases of $\mathbb{F}_2[t]$?
 - ✓ (a) $\{1, 1+t, 1+t+t^2\}$.
 - ✓ (b) $\{1+2t+t^2, 3+t^2, t+t^2\}$.
- ✓ 11. Is $\{1+t^n, t+t^n, \dots, t^{n-1}+t^n, t^n\}$ a basis of $\mathbb{F}_n[t]$?
- ✓ 12. Let $\{x, y, z\}$ be a basis of a vector space V . Are the following sets also bases of V ?
 - ✓ (a) $\{x+y, y+z, z+x\}$
 - ✓ (b) $\{x-y, y-z, z-x\}$
- ✓ 13. Find a basis of $V = \{(a_1, \dots, a_5) \in \mathbb{C}^5 : a_1 + a_3 - a_5 = 0 = a_2 - a_4 = 0\}$.
- ✓ 14. Let $V = \{p(t) \in \mathbb{R}_2[t] : p(0) + 2p'(0) + 3p''(0) = 0\}$. Show that V is a vector space and find a basis for it.
- ✓ 15. Extend the set $\{1+t^2, 1-t^2\}$ to a basis of $\mathbb{F}_3[t]$.
- ✓ 16. Let $u_1 = 1$ and let $u_j = 1+t+t^2+\dots+t^{j-1}$ for $j = 2, 3, 4, \dots$. Is $\{u_1, \dots, u_n\}$ a basis of $\mathbb{F}_n[t]$? Is $\{u_1, u_2, \dots\}$ a basis of $\mathbb{F}[t]$?
- ✓ 17. Construct three bases for $\mathbb{R}^3$ so that no two of them have a common vector.
- ✓ 18. Construct a basis for the real vector space of all $n \times n$ hermitian matrices.
- ✓ 19. Construct a basis for the vector space of all $n \times n$ matrices of trace 0.
- ✓ 20. Show that $W := \{A \in \mathbb{C}^{2 \times 2} : A^* + A = 0\}$ is a real vector space. Find a basis for W .

21. Find bases and dimensions of the following subspaces of $\mathbb{R}^5$:
22. $\{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_1 - a_3 - a_4 = 0\}$.
23. $\{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_2 = a_3 = a_4, a_1 + a_5 = 0\}$.
24. Span of the set of vectors $(1, -1, 0, 2, 1), (2, 1, -2, 0, 0), (0, -3, 2, 4, 2), (2, 4, 1, 0, 1), (3, 3, -4, -2, -1)$ and $(5, 7, -3, -2, 0)$.
25. Determine the dimension of the vector space of all matrices, over $\mathbb{R}$ or $\mathbb{C}$ or both (as appropriate):
- (a) $n \times n$ matrices.
 - (b) symmetric $n \times n$ matrices.
 - (c) skew-symmetric $n \times n$ matrices.
 - (d) hermitian $n \times n$ matrices.
 - (e) upper triangular $n \times n$ matrices.
 - (f) diagonal $n \times n$ matrices.
 - (g) scalar $n \times n$ matrices.
26. Let U and W be subspaces of a finite dimensional vector space. Prove that if $\dim(U \cap W) = \dim(U)$, then $U \subseteq W$.
27. Show that if U and W are subspaces of $\mathbb{R}^9$ with $\dim(U) = 5 = \dim(W)$, then $U \cap W \neq \{0\}$.
28. Let $U = \text{span}\{(1, 2, 3), (2, 1, 1)\}$ and $W = \text{span}\{(1, 0, 1), (3, 0, -1)\}$. Find a basis for $U \cap W$. Also, find $\dim(U + W)$.
29. Let S be a set consisting of n elements. Let V be the set of all functions from S to $\mathbb{R}$. Show that V is a vector space of dimension n .
30. Let V be the set of all functions $f(t)$ having a power series expansion for $|t| < 1$. Show that V is an infinite dimensional vector space.
31. In $\mathbb{R}^3$, what is $\dim(\text{span}\{e_1 + e_2, e_2 + e_3, e_3 + e_1\})$?
32. In each of the following subspaces U and W of a vector space V , determine the bases and dimensions of U , W , $U + W$ and of $U \cap W$.
- (a) $V = \mathbb{R}^3$, $U = \text{span}\{(1, 2, 3), (2, 1, 1)\}$, $W = \text{span}\{(1, 0, 1), (3, 0, -1)\}$.
 - (b) $V = \mathbb{R}^4$, $U = \text{span}\{(1, 0, 2, 0), (1, 0, 3, 0)\}$,
 $W = \text{span}\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 1)\}$.
 - (c) $V = \mathbb{C}^4$, $U = \text{span}\{(1, 0, 0, 2), (3, 1, 0, 2), (7, 0, 5, 2)\}$,
 $W = \text{span}\{(1, 0, 3, 2), (10, 4, 14, 8), (1, 1, -1, -1)\}$.
33. Let V be a vector space. Let $v_1, \dots, v_n \in V$. Show the following:
- (a) If $v_1, \dots, v_n$ span V , then $v_1, v_2 - v_1, \dots, v_n - v_1$ span V .
 - (b) If $v_1, \dots, v_n$ are linearly independent, then $v_1, v_2 - v_1, \dots, v_n - v_1$ are linearly independent.
34. Determine conditions on the scalars a and b so that the vectors $(1, a, 4)$, $(1, 3, 1)$ and $(0, 2, b)$ are linearly dependent.